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RESEARCH METHODOLOGY

Unit - 1



RESEARCH PREPARATION, PLANNING AND RESOURCES

Syllabus :

- Introduction
- Research – objectives, definitions – knowledge economy
- Perspectives of stakeholders
- Types of Research
- Critical Thinking and Creative Thinking.
- Generating Research Topic
- Identifying Broad Research Question and Objectives – Design of Questionnaire
- Characteristics of a Good Research – Research Design
- Methods of scientific enquiry – theoretical, experimental and empirical.
- Hypothesis – Formulation and Testing
- Sources of Information – ICT enabled tools



RESEARCH PREPARATION, PLANNING AND RESOURCES

Syllabus :

- Databases, Repositories, Public and Private Sources, Indexes.
- Literature Search – Keyword
- Back ward and Forward Search
- Quality Measurement Tools – Journal Metrics, Author Level Metrics
- Meta Analysis of Research Findings
- Literature Review – Grouping, Analyzing and Comparison – Identification of Research Methods
- Intellectual Property Rights
- Copyright – Patent – Geographical Indication – Industrial Designs – Technology Transfer.



DEFINITIONS OF RESEARCH

Research :

$$\text{Research} = \text{Re} + \text{Search}$$

- ‘Re’ means again and again and ‘search’ means to find out something.
- The word research is combination of “re” and “search,” which means a systematic investigation to gain new knowledge from already existing facts.
- The following is the process,





- The economy of a country is controlled by several factors, viz., population size, industrial activities, agriculture, policies of its government, culture of the people, educational system, infrastructure facilities, etc.
- In the process of satisfying the basic needs (food, shelter and clothing) people engage themselves in various activities such as agriculture, housing and textile industries.
- Each of the above industries is backed by various other industries.
- For example, a textile industry is supported by machine tool, agriculture and chemical industries.
- Major portion of the product of each of the industries is consumed by the public and a portion of it is fed as raw materials / equipment's to other industries.



- Over and above these industrial activities, education and training of required manpower for these industries are taken care by educational institutions.
- In addition to these activities, there are many service organizations viz., health, transportation, postal, legal, insurance, banking, etc., catering to the needs of the society.
- Policies and guidelines of the central government and state governments facilitate the integration, coordination and control of all the activities of the nation with the objective of maximizing the growth of the nation.
- The above discussion makes it clear that each entity of the nation is forced to have competitive role to maximize its productivity for its survival.
- The productivity of different organizations can be improved through several research studies.



Research :

- It is basically search for truth.
- Research contributes significantly to the progress of the nation as well as an individual with commercial, social, and educational advantages.
- Research is an important parameter to judge the development of any nation.
- Research is an important component of private and government sectors.
- Nowadays, interdisciplinary research is at high demand.
- Research should always aim at providing efficient solutions to routine problems.
- Researchers should carefully choose the appropriate research method and follow a research process by referring to existing theories.



Objectives of Research:

- The main purpose of research is to find solutions to unsolved problems using scientific procedures and also to understand various phenomena scientifically.
- In addition, one of the major objectives of research is to find out a hidden, undiscovered truth.
- There are various objectives behind undertaking research by individuals as well as various organizations.
- Propose and test certain hypotheses that provide fundamental relationships between variables.

Objectives of Research:

- Discover and establish the existence of relationship, association, and independence between two or more aspects of a situation or phenomenon. (Such studies are known as correlational studies)
- Understand different phenomena and develop new perceptions about it
- Test reported findings and conclusions on new data and novel conclusions on previously reported data
- Study the frequency of research that is connected with unspecified study. (This type of study/research is known as diagnostic research)



MOTIVATION BEHIND RESEARCH

- Research is a long process, so the main driving factor is motivation.
- For some researchers and post-graduate students, the main objective behind the research is to earn a degree.
- For organizations including defence and research laboratories, research is an important aspect.
- Initially the research is a random walk (research scholar is not sure about topic/ research problem), but one need to systematically continue to get the destination.
- Failure is an inevitable step in the research phase, for example, failure in getting results, publications, and so on.
- But the researcher's passion and motivation helps in such situations.



IMPORTANT INGREDIENTS FOR RESEARCH

- Creativity, good written and verbal communication skills and in-depth knowledge of the subject are essential for successful completion of research work.





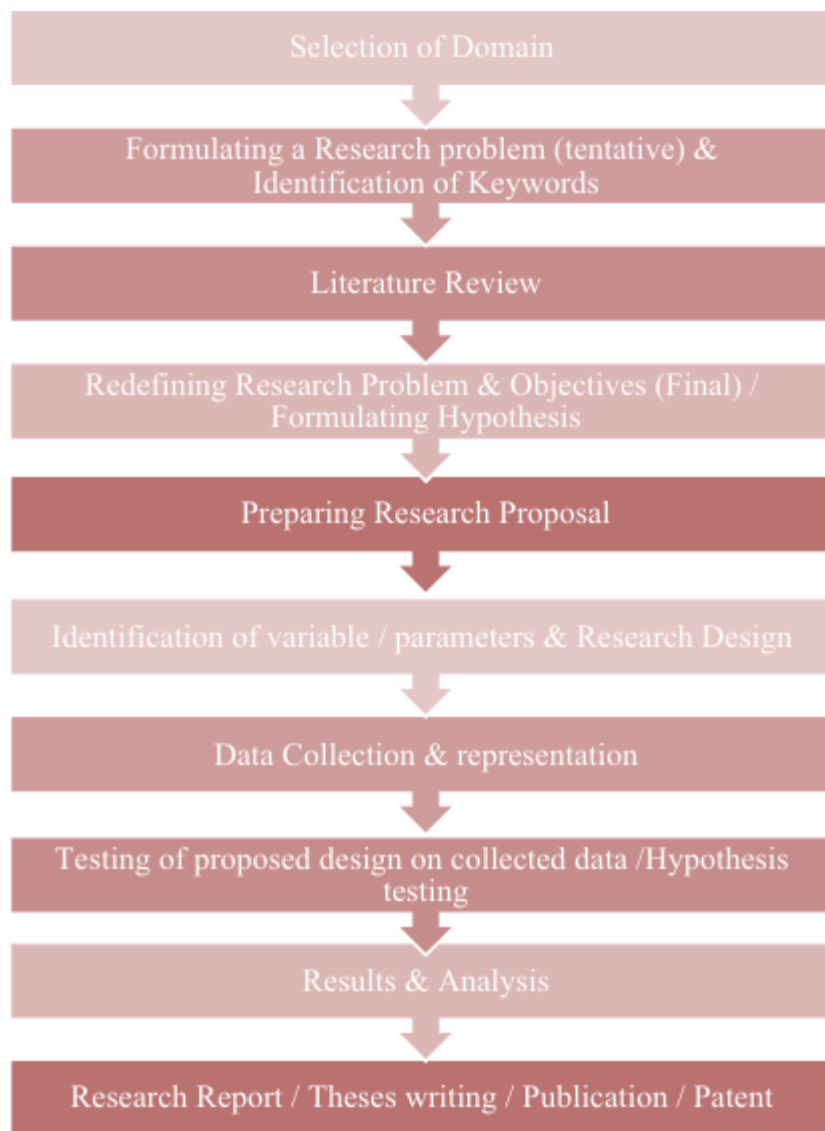
- A researcher should have sound fundamental knowledge of the domain to be undertaken.
- A querying attitude is one of the important factors.
- Anything and everything is questionable; this questioning attitude is essence of research and invention.
- Practical intelligence is the ability to adopt day-to-day requirements.
- One should always challenge conventional solutions to a problem.
- Analyzing a situation in a different way is also creativity



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RESEARCH PROCESS





FEATURE OF A GOOD RESEARCH

- Good research should clearly define the methodologies used, which should be replicable.
- It should be time-bound and realistic.
- Good research should have systematically chosen methodologies and datasets to prove the proposed hypothesis.
- Validity and reliability of data should be checked and researchers should consider an sufficient amount of data.
- Good research should be creative and valid in the longer term.

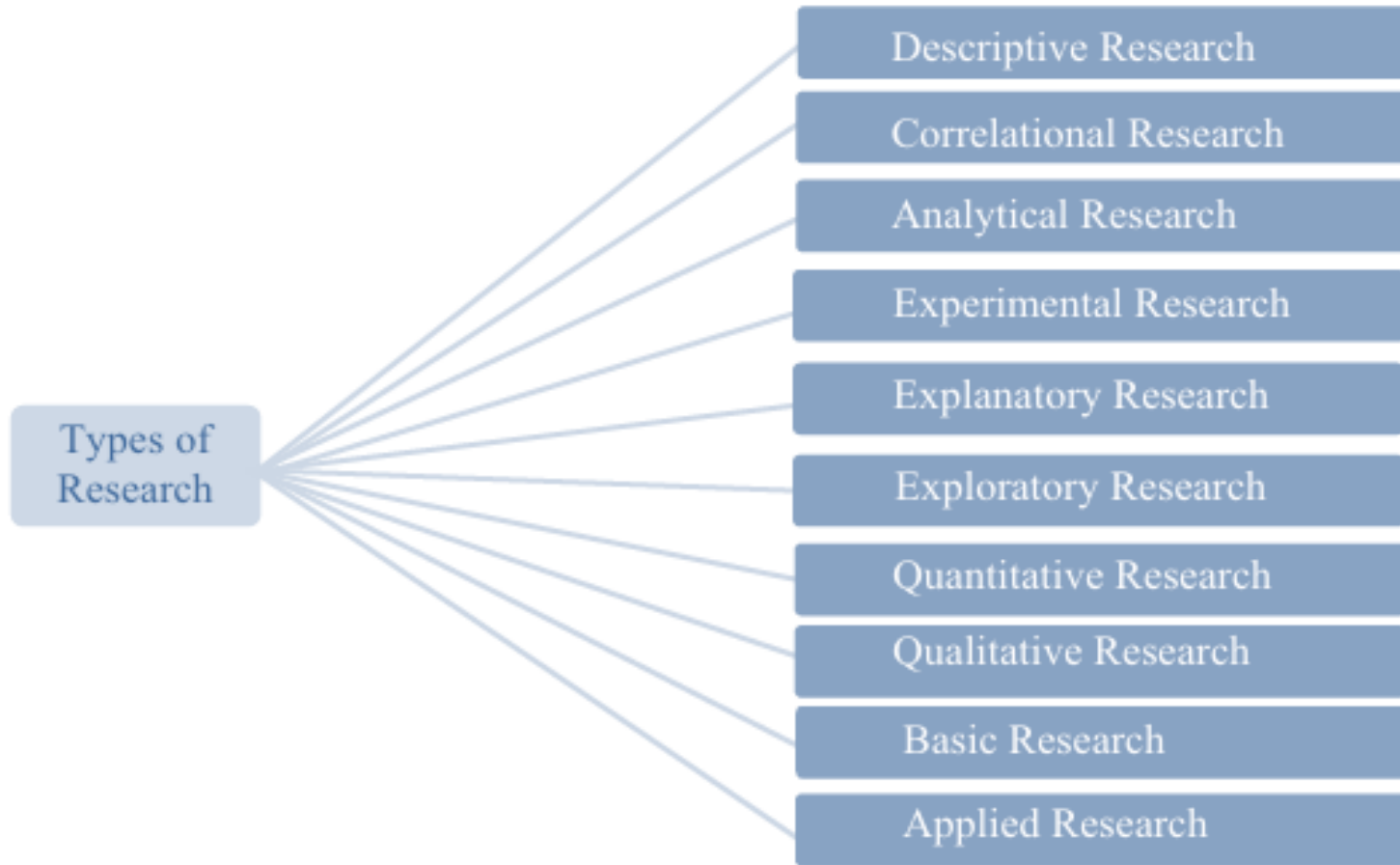


FEATURE OF A GOOD RESEARCH

- Research purpose should be clearly defined
- Procedure for the research should be detailed sufficiently which should help the other to continue the work by referencing our work
- Research work should be carefully planned to get the results in related to the specified objectives
- Reports should be created by a researcher stating that what was the procedure adopted for completing the work which should also include errors in their findings.
- Conclusions should be confident to those justified by the data of research



TYPES OF RESEARCH



TYPES OF RESEARCH AND ITS GOALS

Pure Research (or) Basic (or) Fundamental (or) Theoretical Research.

- There is no immediate need, but new theories can be added to the knowledge cluster.
- This type of research may solve problems but may not have practical applications.
- It has a broader scope compared to applied research.
- Theories in basic sciences and mathematics are examples of basic research.

Example: Newton's laws of motion.

- This has been applied in many product design and testing.

TYPES OF RESEARCH AND ITS GOALS

Applied Research:

- Applied Research tries to solve an immediate specific problem faced by industry or society.
- The obtained solution can be deployed to solve the problem.
- The duration of applied research is shorter as a quick solution is expected.
- The outcome of the applied research either address the unsolved problem or improve the existing solution.

Example:

- “Analysis of cell/body organ behavior in cancer”
- A researcher may use data analytics, image processing, algorithms, and knowledge of the medical industry.

Descriptive Research :

- Descriptive research is generally used in business analysis or social problems.
- This type of research does not have any control over the parameters or variables.
- It just tries to represent or analyze the previous and or current facts.

Example:

- An analysis of customer purchase patterns, that is, purchases from the mall, online, or retailer, as well as the study of travel mode used by people.
- All kinds of correlational methods, survey methods, and comparative studies are descriptive research.

Analytical Research :

- Analytical research uses existing information to explain a complex phenomenon or to perform a critical evaluation.
- The identified hypothesis can be accepted or rejected depending on the analysis; from experience the hypothesis can be redefined.
- Analytical research is observed in historical study, forensic work, food, in the medical domain, and so on.

Correlational Research :

- Correlational research focuses on exploring the relationship or association between incidences, variables, and so on.

Example:

“To study the effect of a modern lifestyle on obesity” :

- A modern lifestyle and obesity are two variables and researchers should study a group of people living both a modern lifestyle and a non modern lifestyle.
- The groups should be segregated on the basis of obesity parameters.
- Collected data may be analyzed to establish the relationship between two variables: “obesity” and “modern” lifestyle.

Qualitative Research :

- Qualitative research mainly deals with the quality or the types of the parameters considered for the research.
- Here, it is assumed that the world is unstable and differences in the parameter may occur with time.

Example:

- **Research related to human behaviour.**
- Everybody can react to the situation differently and it is difficult to propose the predictive conclusions.
- This type of research is more complicated and requires more guidance.

Quantitative Research :

- Quantitative research involves measurements of quantities of characteristics that can be used as features for the research study.
- Statistical quantities that can be measured are involved in quantitative research.
- ***Example:***
- To conduct a survey every weekend and a random question is been asked, “Who is your favorite actor?” and response of this questions is self-controlled by providing a multiple choice question.
- The outcomes are provided briefly in judgment form with statistical respective.

TYPES OF RESEARCH AND ITS GOALS

Experimental Research :

- Experimental research focuses on the fieldwork and experiments that can control the independent variable.
- ***Example:***
- **Medicine Domain:**
- Study of the effect of the new drug on a specific group of people or animal is an example of experimental research in medicine domain.

Explanatory Research (or) Fundamental Research :

- Explanatory research tries to analyze and justify the reason behind the occurrence of particular phenomenon or association between the variables.
- It basically answers the “Why” type of questions.
- It aims to explain why a relationship, association, or inter dependence exist.
- Three important components like time-to-time sequences which will occur before the effect, concomitant variations, where the variations will be systematic between two variables.

Example:

- 1. Why the modernization creates health problem?
- 2. Which company will be more effective and why?



TYPES OF RESEARCH AND ITS GOALS

Exploratory Research :

- This kind of research generally explores the areas that have required attention or it is for checking the possibility of research in the particular domain or area.
- Depending on outcomes of exploratory study, domain is further explored for in depth research on the specific topic.
- Exploratory studies are also conducted to develop, refine and test procedures, policies, and tools.
- ***Example:***
- Why product sales are reduced, due to already existing data or the products which have been acquired recently for the agricultural company?



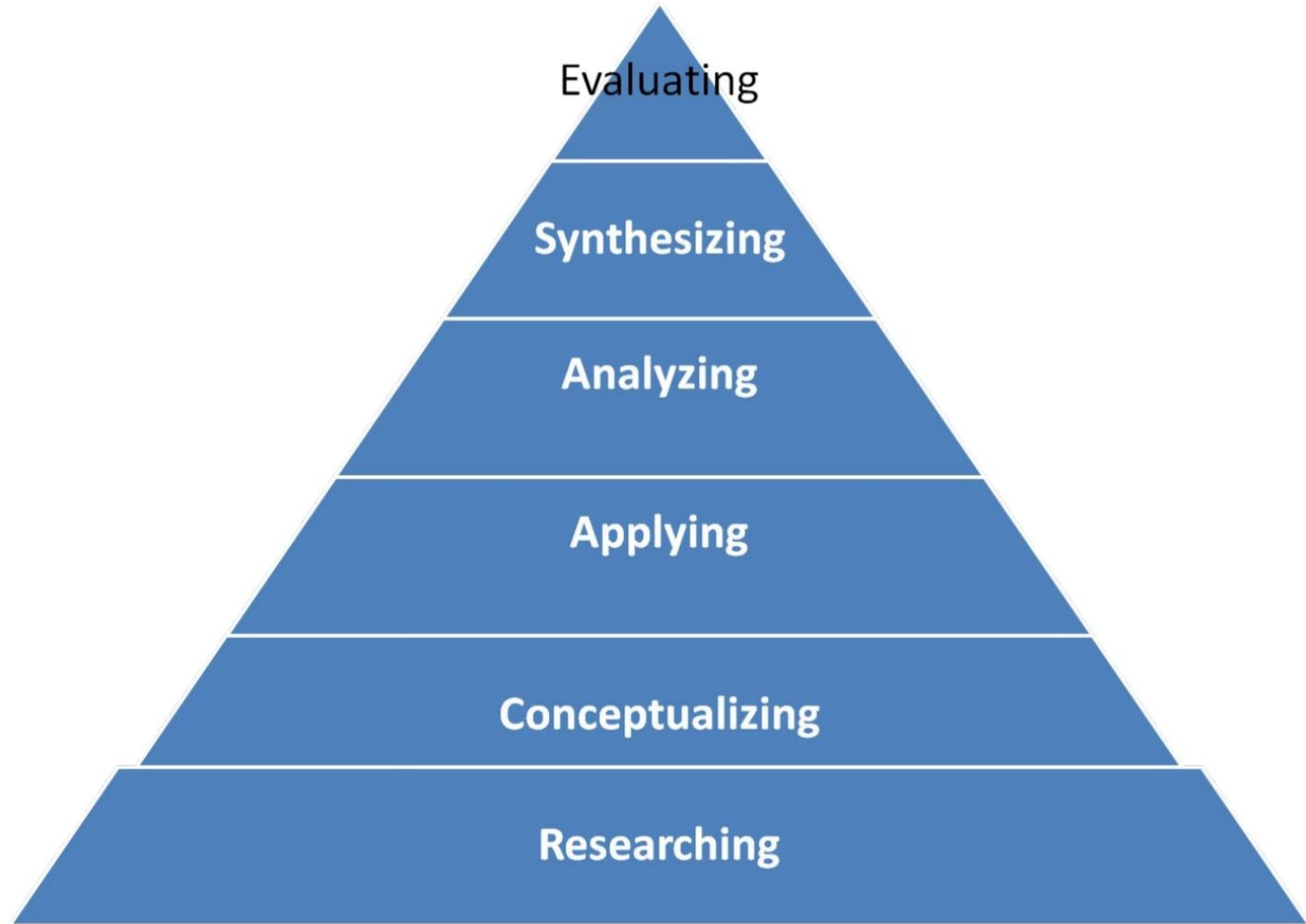
CRITICAL THINKING

- Critical thinking is thinking about thinking in order to make it better
- Critical thinking is a self-directed, self-disciplined, self-monitored and self-corrective thinking process by which we take deliberate steps to think at the highest level of quality.
- The purposeful and reflective judgment about what to believe and what to do in response to the observations, experience, verbal or written expressions or arguments.



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CRITICAL THINKING



Scientific Enquiry:

- The scientific enquiry is a body of techniques for investigating phenomena, acquiring new knowledge, or correcting and integrating previous knowledge.
- To be termed scientific, a method of enquiry must be systematic, controlled and based on empirical and measurable evidence.
- The goal of a scientific enquiry is to obtain knowledge in the form of testable explanations that can predict the results of future experiments.



Objectives of Scientific Enquiry:

- Identify a problem that defines the goal of research.
- Make a prediction that, if confirmed, resolves the problem.
- Gather data relevant to this prediction.
- Analyze and interpret the data to see if it supports the prediction and resolves the question that initiated the research.

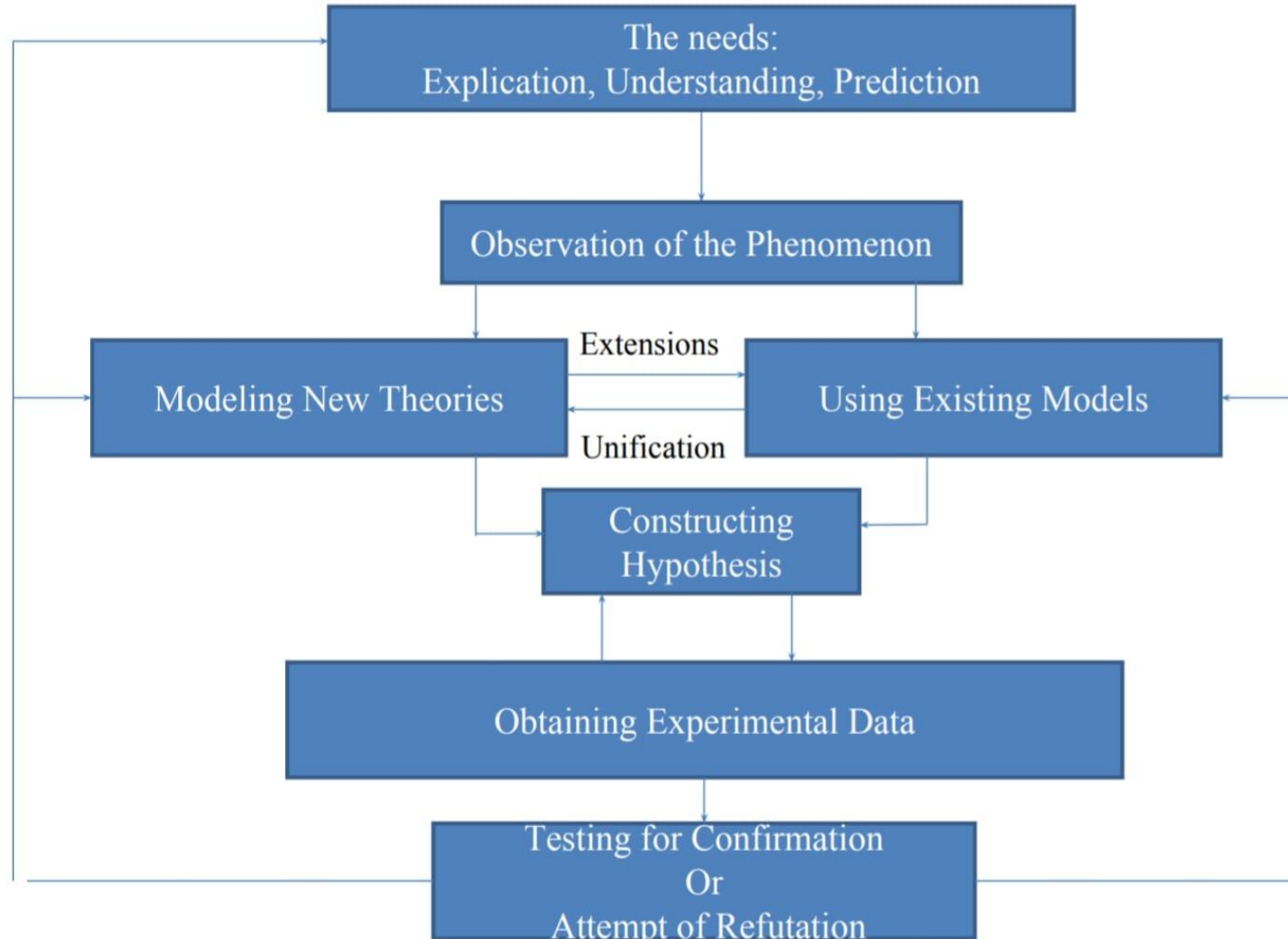


Elements of Scientific Enquiry:

- Characterizations (Observations, definitions, and measurements of the subject of inquiry)
- Hypotheses (theoretical, hypothetical explanations of observations and measurements of the subject)
- Predictions (reasoning including logical deduction from the hypothesis or theory)
- Experiments (tests of all of the above)



STEPS IN SCIENTIFIC ENQUIRY





Designing Questionnaire:

- Characterizations

(Observations, definitions, and measurements of the subject of inquiry)

- Questionnaire – indirect enquiry

- A questionnaire is form containing series of questions and providing space for reply to be filled in by the respondent itself.
- A questionnaire is a formalized set of questions to generate the data necessary for a accomplishing a research project's objectives.

Schedule:

- A schedule is the name usually given to a set of questions which are asked and filled by a interviewer in a face to face to situations.



TEN POINTS TO BE CONSIDERED IN DESIGNING QUESTIONNAIRE

- Must be initially related to final objectives of the investigations
 - Specify the information needed
- Questions must follow logical sequences
 - Arrange the questions in proper order
- Specify the type of interviewing method
 - Biased questions should be avoided
- Proper structuring of the question is must
 - Decide on the question structure
- Should be brief, clear and simple
 - Determine the content of individual questions
 - Determine the question wording

TEN POINTS TO BE CONSIDERED IN DESIGNING QUESTIONNAIRE

- Reproduce the questionnaire
 - Question should have appropriate and to it
- Eliminate bugs by pretesting
 - Conflict between interviewer and respondent should be avoided
- Questions outside the respondents experience should not ask
 - To overcome the respondent's unwillingness to answer
- Questions should not touch memory of the respondent
 - To overcome the respondent's inability to answer
- Identify the form and layout
 - To gather factual data

What is a research problem?

- Research problem refers to some difficulty / need which a researcher experiences in the context of either theoretical or practical situation and wants to obtain a solution for the same
- The first step in the research process – Identification / selection and formulation of the research problem

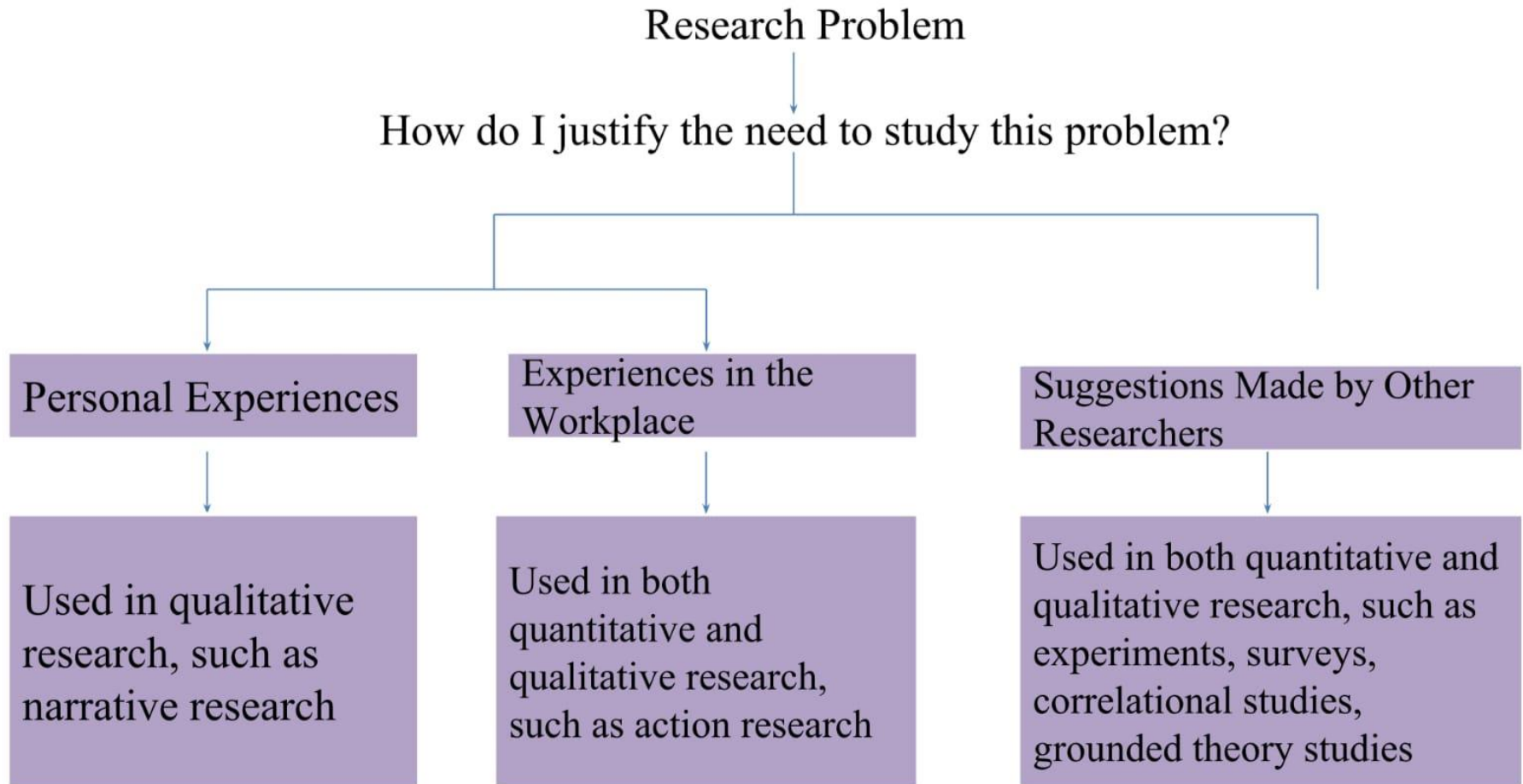


RESEARCH PROBLEM FORMULATIN - STEPS

1. Statement of the problem in a general way
2. Understanding the nature of the problem
3. Surveying the available literature
4. Developing ideas through discussions
5. Rephrasing the research problem



PROCESS OF JUSTIFYING A RESEARCH PROBLEM





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- Study population – they provide information or researcher collect information about them
- People – Individuals, organizations, groups, communities
 - Subject area – information that researcher need to collect to find answers to research question
- Problems – issues, situations, associations, needs, profiles
- Program – content, structure, outcome, attributes, satisfactions, consumers, service providers, etc.
- Phenomenon – cause and effect relationships, they study of phenomenon itself

TOPIC SELECTION CRITERIA

Internal / Personal criteria

- Researcher's interest and inclination must be matched by competence before researcher can decide the area of research.
- Competence can be cultivated and learning new techniques is a part of any research program

External criteria or factors

- Researchability of the problem, Importance and urgency, Novelty of the Problem, Feasibility, Facilities, Usefulness and Social Relevance, Research Personnel

Rational thinking and Creative thinking are recommended along with above criteria



TOPIC SELECTION CRITERIA

Points to be considered in Topic Selection

1. Once the main area of research is decided upon the next step is to narrow down the topic
2. Research supervisor has a greater role to play
3. Acquaintance with the field – books / dissertations, journal, papers, etc.
4. Reading dissertations to avoid duplication of work
5. Sincere efforts to be made for a general survey of literature before narrowing down the topic
6. Well informed research supervisor can afford real guidance to the researcher



HYPOTHESIS

What is a Hypothesis?

- A hypothesis is written in such a way that it can be proven or disproven by valid reliable data**Griennell & Stothers**
- A tentative statement about something, the validity of which is usually unknown.**Black & Champion**

Example:

- The drug is not effective in curing the disease (or) the drug is effective in curing the disease.

What are the characteristics of the hypothesis?

- A hypothesis must be formulated in simple, clear and declarative form
- A hypothesis must be empirically testable
- A hypothesis must be measurable and quantifiable so that statistical analysis can be established
- The validation of hypothesis would necessarily involve testing the statistical significance ($\alpha=0.05/\alpha=0.01$) of the hypothesized relation



IMPORTANT ASPECTS OF A HYPOTHESIS

- A hypothesis should be simple, specific and conceptually clean
- A hypothesis should be capable of verification
- A hypothesis should be related to the existing body of knowledge
- A hypothesis should be operation able

Null Hypothesis(H_0) :

There is no significant difference between the sample statistic and the corresponding population parameter or between two sample statistics is called Null Hypothesis.

Alternate Hypothesis(H_1) :

A hypothesis that is different from the null hypothesis.

Two Tailed : $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$,

One Tailed : $H_0 : \mu = \mu_0$, $H_1: \mu > \mu_0$ (***Right Tailed***)

$H_1: \mu < \mu_0$ (***Left Tailed***)

HYPOTHESIS

Test of Significance : A very important aspect of the sampling theory is the study of tests of significance which enable us to decide on the basis of the sample results if

- (i) The deviation between the observed sample statistic & the hypothetical parameter value is significant.
- (ii) The deviation between two sample statistics is significant.

Type I Error : Reject H_0 when it is true. **Type II Error :** Accept H_0 when it is wrong.

Level of significance (α): In testing a given hypothesis the maximum probability with which we would be willing to take risk is called the level of significance of the test

- The **researcher should be very careful** in interpreting the results after applying tests of significance
- The success of the statistical test depends on the availability of **correct and reliable data**



HYPOTHESIS TESTING PROCEDURE

1. Set up the null hypothesis H_0 .
2. Set up the alternative hypothesis H_1 .
3. This will enable us to decide whether we have to use a single tailed (right or left) test or two tailed test.
4. Choose the appropriate level of significance (either 5% or 1% level). This is to be decided before sample is drawn.
5. Compute the test statistics $z = \frac{t - E(t)}{sE(t)}$ under the null hypothesis.
6. We compare the computed value of z in step (4) with the tabulated value z_α at given level of significance α .
7. If the calculated value of Z is less than tabulated value z_α then H_0 accepted.
8. If the calculated value of Z is greater than tabulated value z_α then H_0 rejected.



Literature Survey :

- The literature survey researcher has to first define tentative research problem definition and identify the related keywords for literature search.
- The literature survey is a comprehensive study of technical and authorized content related to research keywords.
- It is very important step in the initial phase of research, however, this step is revisited by researcher number of times during research journey.
- Literature survey provide details of research progress of particular domain.
- It helps the researcher to understand the approaches, methodologies, algorithms, and datasets used by other scientists.



Literature Survey :

- Also, it is important to identify where the gap is.
- Where can I contribute to an existing knowledgebase?
- It helps the researcher to understand the progress of domain and state of the art in the domain.
- Literature survey also helps to avoid duplication of work.
- One should always avoid reinventing the wheel; this is important if one of the purposes of the research is to get a PhD degree.



Literature Survey :

- Open Access journals are one of the good sources of research articles, one can also access articles from library subscription, some university libraries etc.
- Good literature sources are periodicals, conference proceedings, research reports, standards, theses, dissertation reports, research reports, patents, reviews, textbooks, hand books, encyclopaedia etc.
- There are different strategies to search literature.
- One needs to first identify the keywords related to the problem definition and searching articles using keywords is the more precise way to search article.



Backward and forward reference search is also popular and beneficial to understand the background of the research area.

Backward Literature Survey :

- One can choose the latest article from the reputed journal written by good researcher and study all related references of that paper. This is a backward reference.

Forward Literature Survey :

- One can also find the important old article in the domain and find all the papers that referred the article. This is forward reference search.



The Beginner of the researchers can follow below steps for literature survey:

- (i) First download 100–120 research papers from published papers of IEEE / ACM / Wiley / Elsevier / Springer or some other good SCI Indexed Journals related to research topic and start reading it.
- Normally, we do this type of search with the help of keywords of research problems.
- Some of the search engines for the literature survey are Google Scholar, CiteSeer, and so on.



The Beginner of the researchers can follow below steps for literature survey:

- (ii) Researchers can initially read only title, abstract and conclusion of each research paper and decide whether it is related to research topic.
- If yes, include in list, else skip it.
- From this step you will get to know “What has been done in this area and what can be done in future?”
- From this understanding one can come up with his/ her research problem statement.



The Beginner of the researchers can follow below steps for literature survey:

- (iii) Read shortlisted papers in detail and start taking notes of different parameters considered, methodologies used, conclusions derived, and related mathematics/ theories used.
- Researchers can finalize research hypothesis after this step and also can write good survey paper.



The Beginner of the researchers can follow below steps for literature survey:

- Another approach for literature survey is selecting any one best paper related to particular research problem of your interest.
- Try to find/download those papers that are cited and included in the references section.
- This approach leads to a better literature survey of a particular research problem.
- During the literature survey process, researchers should able to identify the top 10 researchers in his/her domain.
- He/she should find good universities where people are working on the same or similar problems.



The Beginner of the researchers can follow below steps for literature survey:

- Literature survey is a step that is often visited during the research journey.
- Researchers should always look at what other top researchers are doing in respective area of research.
- Researchers should provide main focus on comparing the approaches, Identifying the weaknesses and strengths in recent research articles in the subject.
- Each paper's study can be summarized in systematic manner.



The Beginner of the researchers can follow below steps for literature survey:

- Literature surveys can be done with respect to following points:
 - What
 - How(process schematic)/standard procedure (system block schematics)
 - Discussion on major steps involved
 - Design criteria and performance measures
 - Techniques currently in use
 - Comparative analysis (table or any suitable tool to discover and list pros and cons/strength/weakness/future scope of existing techniques)
 - Scope for research/gap in research



The Beginner of the researchers can follow below steps for literature survey:

- Researcher may add some more parameters to be studied about the research paper and can prepare a format for same to keep record of each paper.
- Summarization and representation of literature surveys is an important part in the theses, presentations and in journal writing.
- Literature survey can be represented in textual, tabular or graphical format using different parameters.



- **The survey of literature and review of literature** is an integral part of the research to acquaint with the existing knowledge or material available in the area of interest.
- Literature survey means penetrating through the available literature in the selected area of research.
- The literature review involves assessment and examination of the reported contributions/findings.
- Recording, summarizing and evaluating the existing findings and being able to provide expert judgment of the methods and finding is also called as literature review.



Example:

- When you are travelling from one city (source) to the other city (destination), consider two possible ways of finding your destination.
- **In the first case one can go by railway.**
- In this case, there is no need to know the route from the source to destination.
- **Second way is to use your own car.**
- In this case, you need to take help of geographical map (i.e. supervisor/guide in research).
- In Google maps, various routes are available such as low traffic route, fastest route, shortest route and so on.
- You need to plan your journey to reach to the destination.



Example:

- In the same way, when you are pursuing your research, you need to plan your journey of research, you need a help of your guide (like a Google map).
- Also, you need to devise your own plan to reach the destination.
- Literature survey helps you to find out many answers while pursuing the research.
- It can be time consuming in initial stage.
- But if literature survey is not done in initial stage, then you might get stuck up while pursuing the research.
- Some authors have also given the meaning of literature review as the critical study of literature survey.



- A researcher cannot progress in the area of interest without the literature survey.
- A literature survey can be divided into three parts
 - Searching of the literature in the area of interest
 - Collection/acquisition of literature or data
 - Critical study of literature/review of literature in the area of interest
- Literature review helps you to widen the knowledge in the selected domain.
- It is very important to know the contributions of other researchers in the existing area or domain.



- Moreover, questions like, what challenges, which problems, what advantageous and disadvantageous of the existing systems, should trigger the thought process.
- These questions help to focus on finding answers to those limitations and accordingly you can define your own problem statement.
- Literature survey and literature review should go hand in hand.
- After searching the material available in the selected area of research, researcher has to read it, understand it, and note down the important points.
- Again, find the latest literature, read it, understand it, and note important points.
- This process of literature survey and literature review should be continuous.



Literature Review :

- A literature review is a written summary of journal articles, books, and other documents that describes the past and current state of information on the topic of your research study.

1. Objectives of Literature Review:

- Discovering important variable
- Distinguishing what has been done from what needs to be done
- Synthesizing the available studies to have perspective
- Determining meanings, relevance of the study and relationship with the study and its deviation from the available studies.



2. Sources of Information

- **Primary source literature** – reported by individual(s) who actually conducted research
 - Research articles, patents, photographs, letters.
- **Secondary source literature** – literature that summarizes primary sources (basic findings)
 - Monographs, histories, reviews, bibliographies.
 - Meta-analysis – statistically combine or compare research results.
- **Tertiary source literature** – synthesizing information gathered from other resources
 - Reference back to primary / secondary sources
 - Handbooks, encyclopedias, magazines, newspaper articles, lectures.



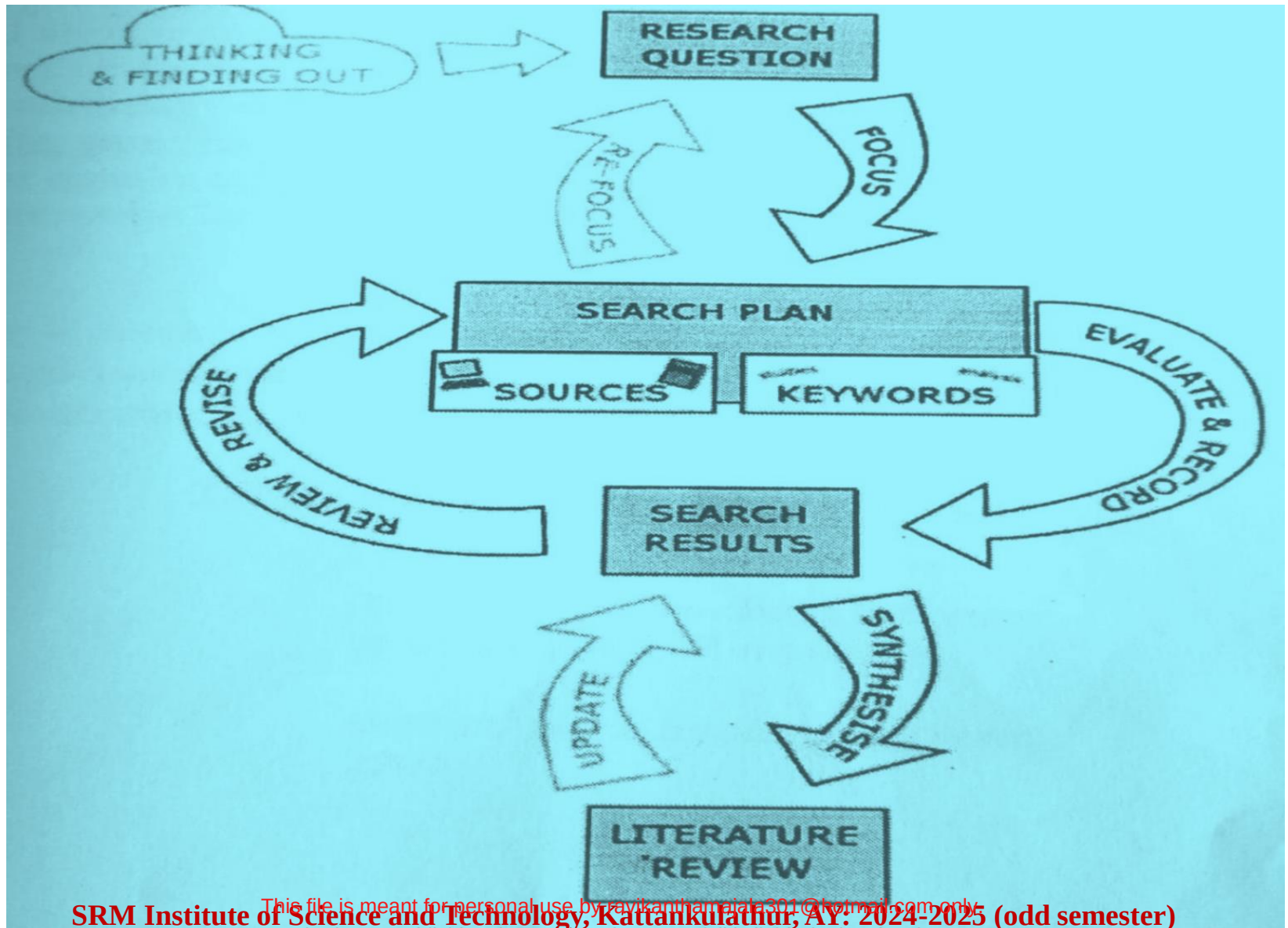
3. Literature Search

- A literature search is a comprehensive survey of publications and information on a specific topic
- Literature search approaches:
 - Systematic – you try to find all relevant material
 - Retrospective – you find the most recent material and work backwards
 - Citation – you follow up references from useful articles, book and reading lists
 - Targeted – you restrict your topic and focus on a narrow area of the literature
- The result produced at the end of a literature search is usually a list of references



4. Five steps in conducting a literature review

- (i) Identify key terms
- (ii) Locate literature
- (iii) Critically evaluate and select the literature
- (iv) Organize the literature
- (v) Write a literature review (Record of research review)





(i) Identify key terms

- Write a preliminary “working title” for a project
- Pose a short, general research question
- Use words that authors report in the literature
- Look in a catalog of terms to find words that match your topic
- Bookshelves in a college or university library

(ii) Locate literature

- Use academic libraries
- Use primary, secondary and tertiary sources
- Search different types of literature – Online databases, journals and early stage literature



(iii) Critically evaluate and select the literature

- It is a good, accurate source?
- Is it relevant?
 - Topic relevance
 - Individual and site relevance
 - Problem and question relevance
 - Accessibility relevance



(iv) Organize the literature

- Priority system for literature search:
 - Refereed journal articles > Non-refereed journal articles > Books
> Conference papers > Dissertations and theses > Non-reviewed articles in websites
- Reproducing, downloading, and filing:
 - Make copies, scan or download the articles
 - Place articles in file folders (also store them in computer file)
- Organize by author name, sources, topic, or key words
- Taking notes and abstracting studies:
 - Note taking – informal procedure
 - Abstract – summary of the major aspects of a study or article (not more than 350 words)



- Constructing a Literature map:
 - A literature map is figure or drawing that displays the research literature on a topic
 - Used to determine how a proposed study adds to the existing literature

(v) Write a literature review (Record of research review)

- Report summaries of the literature for inclusion in your research report.
- Should include introduction, summary and critique of journal articles, justifications for your research project and hypothesis for your research project.
- Style manual provides a structure for citing references, labeling headings and constructing tables, figures for research report



Review Report Writing Approaches

- **Topical order** – organize by main topics or issues
- **Chronological order** – organize by dates the research was published
- **Problem-cause-solution order** – organize by moves from problem to the solution
- **General-to-specific order** (Funnel approach)-examine broad-based research first and then focus on specific studies that relate to the topic
- **Specific-to-general order** – discuss specific research studies so conclusions can be drawn



(i) Online Data Bases:

- Database – collection of records
- Online database – collection of data presented in a searchable format via online
 - Web-based electronic indexes to locate and retrieve articles, magazines, journals and newspapers.
- Types of online database
 - Multidisciplinary (comprehensive) vs. subject specific
 - Full text vs. citation only



(ii) Six Important Online Databases:

1. ERIC database – www.eric.ed.gov-Free access
2. Psychological abstracts – www.apa.org-PsycLit, PsycINFO, PsycARTICLES
3. EBSCO Information services-www.ebsco.com
4. Social Science citation Index (SSCI) from ISI
5. Sociological Abstracts
6. Dissertation Abstracts



Citation Indices

- A citation index allows you to determine the research impact of your publications according to the number of times it has been cited by other researches
- Institute for Scientific Information (ISI) citation indexes are available online to users via ‘Web of science’ system

(i) Citation Indices: invented by Eugene Garfield

- “An ordered list of cited articles each of which is accompanied by a list of citing articles. The citing article is identified by a source index, the cited article by a reference citation. The reference is arranged by reference citations”.



(ii) Five types of citation indices:

- **Science Citation Index (SCI) Expanded** – 1899 to present
 - A multidisciplinary database, with searchable author abstracts, covering the journal literature of the sciences.
 - It indexes more than 6,650 major journals across 150 scientific disciplines.
- **Social Science Citation Index (SSCI)** – 1898 to present
 - Covering journal literature of the social sciences
 - It indexes more than 1,950 journals spanning 50 disciplines
- **Arts & Humanities Citation Index (A&HCI)** -1975 to present
 - Covering the journal literature of the arts and humanities
 - It indexes more than 1,395 of the world's leading arts and humanities journals.



- **Conference Proceedings Citation Index – Science (CPCI-S) – 1991 to present**
 - Lets you use cited reference searching to see the full impact of conferences and other professional meetings
 - With a global coverage of over 110,000 conference proceedings
- **Conference proceedings Citation Index – Arts & Humanities (CPCI-A&H) – 1991 to present**
 - This citation index covers conference literature in all fields of social sciences, arts and humanities.



Impact Factor

- Devised by Eugene Garfield
- Journals are ranked within category by their impact factor
- Impact factors are calculated yearly starting from 1975 for those journals that are indexed in the Journal
- Citation Reports (JCR)

Principles Underlying Impact Factor

- Impact factor is a measure of frequency with which the ‘average article’ in a journal was cited in particular year.
- Impact factor is calculated based on a three-year period, and can be considered to be the average number of times published papers are cited up to two years after publication.



Computation of impact factor

- **The Impact Factor Algorithm**

- Number of citations in year y to items published in years $y - 1$ and $y - 2$
- Number of citable items published in years $y - 1$ and $y - 2$

- **The 2007 Impact Factor**

- Number of citations in 2007 to items published in 2006 and 2005
- Number of citable items published in years 2006 and 2005



Journal impact factor

$$IF_y = \frac{\text{Citations}_y}{\text{Publications}_{y-1} + \text{Publications}_{y-2}}.$$

For example, *Nature* had an impact factor of 41.577 in 2017:[9]

$$IF_{2017} = \frac{\text{Citations}_{2017}}{\text{Publications}_{2016} + \text{Publications}_{2015}} = \frac{74090}{880 + 902} = 41.577.$$



Immediacy Index :

Cites in 2007 to articles published in 2007 = 2343

Number of articles published in 2007 = 366

Calculation : $\frac{\text{Cites to current articles}}{\text{Number of current articles}} = \frac{2343}{366} = 6.402$

h-index (Hirsch index):

- H-index is a citation measure of the impact of a particular author rather than a journal
- Combines an assessment of both quantity (number of papers) and quality (impact, or citations to these papers)
- h – index is automatically calculated : by ‘Web of Science’, ‘Scopus’ and ‘Publish or Perish’
- h – index can also be manually calculated for an author based on the number of papers authored and the number of times each paper has been cited



Merits of impact factor :

- Publish in high impact factor journal:
 - Higher citations rate means your article has higher chances of getting cited
 - Higher reading rate means your article gets read by more researchers
 - More recognition in terms of scholarly contribution
- Tell us how frequently the average article in a journal has been cited in a particular year
 - Impact factor >1 implied a journal is frequently cited
- Tell us something about a journal as a whole
- In library, it is a tool for management of library journal collections
- In market research, it is a quantitative evidence for publishers to position their journals



Demerits of impact factor

- Is an average; not all articles are equally well-cited
- Only includes “citable” articles in the denominator of the equation, i.e., articles and reviews
- Only a limited subset of journals is indexed by ISI
- It is expensive to subscribe to the JCR
- Biased toward English – language journals
- Tell us NOTHING concrete about any specific paper or specific author



Review articles:

- Review articles are an attempt to summarize the current state of understanding on a topic
- Analyze or discuss research previously published by others
- Review articles will teach us about
 - main people working in a field
 - recent major advances and discoveries
 - significant gaps in the research
 - current debates
 - ideas of where research might go next



Types of review articles: 3 major types

- **Types of methodological approach:**

- Narrative review – author's experience, existing theories and models
- Systematic review (meta-analysis) - finding from various individual studies
- Best evidence review – selection and result exploration

- **Types by objective**

- Status quo review – presentation of most current research
- History review – development of a field of research over time
- Issue review – investigation of an issue
- Theory / Model review – introduction of a new theory or model

- **Types by mandate :**

- Invited reviews – experienced researchers are invited
- Commissioned reviews – formal contracts of authors with clients
- Unsolicited submissions – develop an idea for a review and submit it to journal editors.



Organization of review article

- Title – in this cases does not indicate that it is a review article
- Abstract – includes a description of subjects covered
- Table of Contents – shows the reader the organization of the text (overview)
- Introduction includes a description of context (Paragraph 1-3), motivation for review(paragraph 4, sentence1)&defines the focus(paragraph 4, sentences 2-3)
- Body – structured by headings and subheadings
- Conclusion – states the implications of the findings and an identifies possible new research field
- References (“Literature Review”) – organised by number in the order they were cited in the text.



Advantages of review articles

- Audiences of review articles
 - Experts in specific research areas
 - Students or novice researches
 - Decision – makers
- To organize literature
- To evaluate literature
- To identify patterns and trends in the literature
- To synthesize literature
- To identify research gaps and recommend new research areas



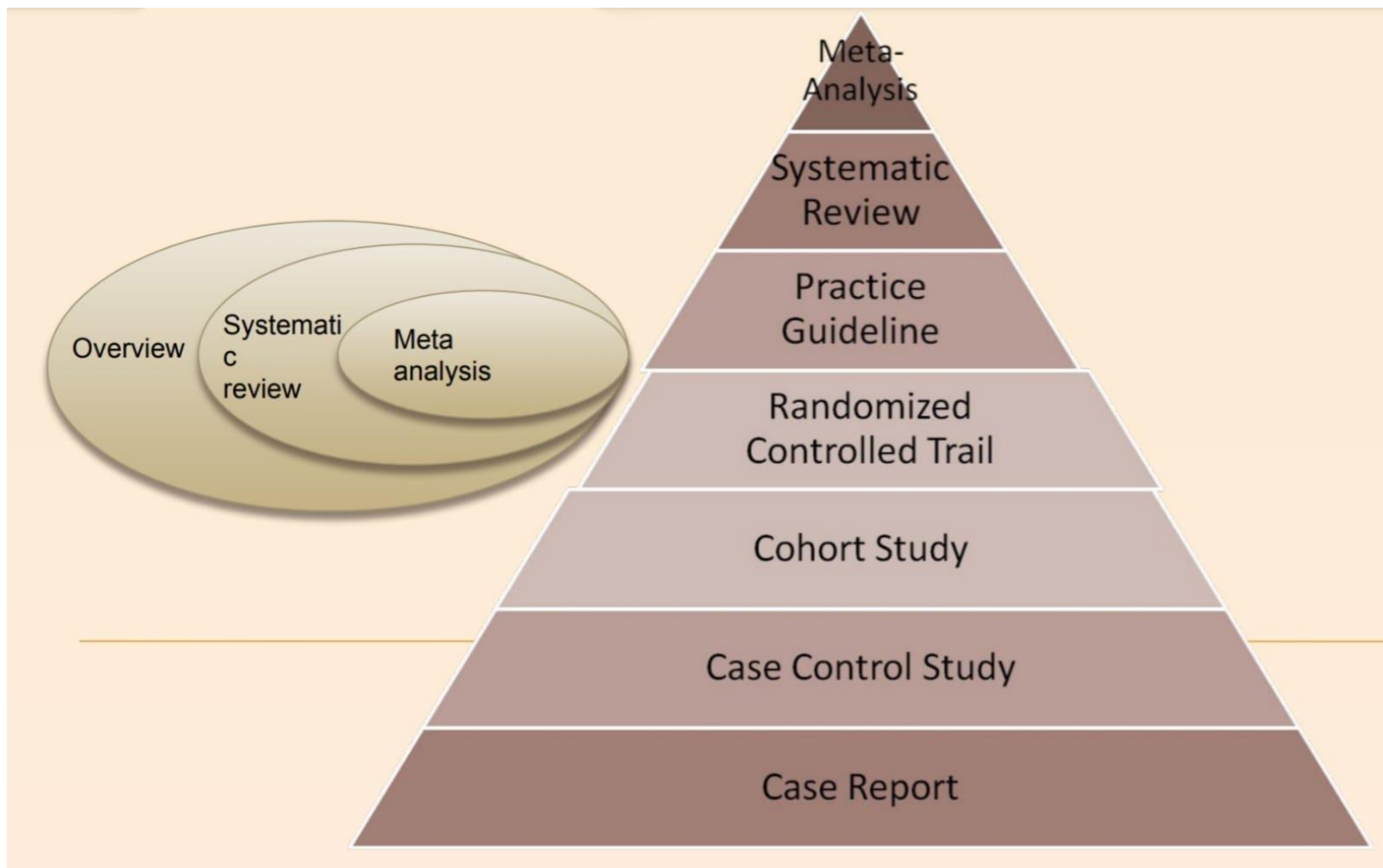
Meta analysis: ‘an analysis of analysis’

- Coined by Gene. V. Glass in educational research
 - Meta-analysis is quantitative approach for systematically combining results of previous research to arrive at conclusions about the body of research.
 - Quantitative: numbers
 - Systematic: methodical
 - Combining: putting together
 - Previous research: what’s already done
 - Conclusions: new knowledge (stronger than the analysis of any single study)



META ANALYSIS

Meta –analysis is used when you are comparing multiple studies that looked at similar relationships





Advantages of meta-analysis

- Results can be generalized to a larger population
- Precision and accuracy of estimates can be improved as more data is used
- Inconsistency of results across studies can be quantified and analyzed
- Hypothesis testing can be applied on summary estimates
- Moderators can be included to explain variation between studies
- Presence of publication bias can be investigated



Ethical and moral issues in Research

- Ethics before research begins
 - Obtain consent, define the ‘gatekeeper’, no data distortion in crafting research methods
- Ethics during research
 - Field notes, consent issues, content issues, moral issues
 - Disposal of data, freedom of information, reuse of data, difficulties of contacting original respondents



Ethics, consequentialism and Misconduct in Research

- Ethics involves the application of fundamental ethical principles to a variety of topics involving scientific research
- Consequentialism refers to those moral theories which hold that, a morally right action is one that produces a good outcome, or consequence
- Misconduct refers to the falsification practices that seriously deviate from those that are commonly accepted honest errors



Ethical issues concerning research participants

- Potential harm to participant – hazardous experiments, discomfort, anxiety, harassment, invasion of privacy, demeaning or dehumanizing procedures
- Code of ethics should be given to everyone who takes part in the research.

This should include

- Autonomous, Dignity, Honesty, Not misrepresenting anyone-written/verbally
- Confidentiality, Data Protection
- Transparency, Openness – Right to comment
- Anonymous, Informed consent
- Collecting information – seeking sensitive information
- Providing incentives



Ethical issues relating to the researcher

- Avoiding bias – Bias is a deliberate attempt to either to hide what you have found in your study, or highlight something disproportionately to its true existence
- Using inappropriate research methodology
 - Ontological lens – what we believe in
 - Epistemological lens – what we accept as valid knowledge
 - Axiological lens – what our personal values are
 - Conceptual framework – how we conceptualise our problem / research issues



Scientific misconduct (academic scandal)

- Plagiarism – cheating
- Using someone else's work without giving them credit for the information
- Fabrication and Falsification – fudging
- Making up or altering research data
- Lazy writing – flaw in writing
- Using too much quoted (although properly cited) material in a report

Whistleblowing

- Exposure to public of wrongdoing by an organization usually by an employee
- Reporting of wrongdoing within part of an organization to senior management



PLAGIARISM

- Plagiarism is a form of intellectual theft.
- Plagiarism is an inappropriate act of claiming credit or ownership of someone else's ideas or unfair annexation of another's work and entitling it as your own.
- Plagiarism includes intentionally copying lengthy passages from a book or journal article, or purchasing or downloading whole papers and submitting them as your own work leading to misconduct.



Types of Plagiarism:

- Plagiarism of Words
- Stitching Sources (Potluck Plagiarism)
- Patchwork Plagiarism
- Self-Plagiarism (the Self-Stealer)
- Cyber and Digital Plagiarism
- Accidental Plagiarism
- Plagiarism of Authorship
- Plagiarism of Ideas
- Reuse of Programming Code



Software Used for Identifying Plagiarism

- After writing any paper or story, it's important to check the paper is free from plagiarism or not.
- Plagiarism free paper is very beneficial for representing quality of work.
- The testing of plagiarism is done by software.
- There are many software tools are used to testing plagiarism.
- The software available online is of two types
 - Paid software (iThenticate, Turnitin, Urkund)
 - Free software



Software Used for Identifying Plagiarism

Paid software (iThenticate, Turnitin, Urkund)

- **(i) iThenticate:**
- iThenticate is very famous software widely used in famous publications IEEE, Elsevier and Springer and so on.
- iThenticate also provide different file type that we can upload easily without any restriction of file type.
- The file types that supported to upload in iThenticate are Microsoft Word® (DOC and DOCX), Word XML, Plain Text (TXT), Adobe PostScript®, Portable Document Format (PDF), HTML, Corel WordPerfect® (WPD) and Rich Text Format (RTF).



Software Used for Identifying Plagiarism

- **(ii) Turnitin:**
 - Turnitin is widely used commercial internet based plagiarism software tool for plagiarism detection services.
 - University and many more IPR filling organizations buy this tool for plagiarism detection.
 - Turnitin checks the submitted document in database and gives the similarities with existing sources.
- **(iii) Urkund:**
 - Urkund plagiarism software is widely used in throughout the Europe for plagiarism check.
 - This software is completely automated used at universities and university colleges.



Free Software

- TO avoid the copyright and plagiarism there are many more free software tools available on internet.
- The most popular tools that are used widely for short plagiarism check:
 - Dupli Checker
 - Copyleaks
 - PaperRater
 - Plagiarisma
 - Plagiarism Checker
 - Plagium
 - PlagScan
 - PlagTracker
 - Quetext
 - Viper



Techniques to Avoid Plagiarism

- One of the essential components that need to be inevitably cultivated and inculcated in human beings is a practice to take pride in your own work and enjoy genuine satisfaction in your own work with real level of attainment matching your actual ability.
- Beyond being unethical, unfair, or dishonest, Plagiarism of any kind, whether caught or not, is always cheating yourself and letting yourself down and should therefore be avoided.
- Citation
- Referencing
- Bibliography
- Paraphrasing



INTELLECTUAL PROPERTY RIGHTS (IPR)

- There are various forms of invention, creativity, research, innovations, artistic creations that are covered by Intellectual property.
- In daily life, we come across various forms of IP like, literary work (e.g., Book), electronic devices (e.g., Mobile), songs, movies, pharmaceutical products etc.
- The legal rights covering IP are called as Intellectual property rights.
- For example, an original literary work and an invention are intellectual property and when protected by the IPR, they will be termed as “copyright” and “patent,” respectively.



INTELLECTUAL PROPERTY RIGHTS (IPR)

- There are two types of properties:
- **1. Physical Property:**
 - It is the tangible property such as Land, Ornaments, House, Money, Building, and so on.
- **2. Intellectual Property:**
 - It is the intangible property such as Patent, Copyright, Trademark, Design, Geographical indications, and so on.
 - Industrial Property - patents, designs, trademarks, trade secrets, and so on.
 - Intellectual Property - copyright for literary works, artistic works, software.
 - Intellectual Property is about creative ideas, such as inventions, music, design, article, poems, and so on.



Copyright Laws vs. Patent Rights

Copyright	Patent rights
Protects expression, not ideas	Protects ideas that have been reduced to practice
Protects the use or performance of original works of literature, art, music, drama, or any other form of expression	Protects the use manufacture or sale of inventions
Life of author + 70	20 years from filing
Covered by 'Copyright Act of 1909', 'Fair Use Doctrine' and 'Digital Millennium Copyright Act (DMCA)'	Covered by 'Federal Law (Title 35 USC)'
Fair Use	No Fair Use, No Research Exemption
Works for hire	Employee inventor owns
Protects against copying	Innocent Infringement not a defense
Protection is automatic (registration is relatively simply)	Extensive examination process before any rights granted
Originality (low bars)	Novelty and Nonobvious (high bars)
In software, copyright automatically covers the authorship in the source code	Patentability standard for software patents is strict and uncertain



Patent:

- A patent is defined as a set of exclusive rights granted by a sovereign state or intergovernmental organization.
- It is granted to an inventor or assignee for a limited period of time in exchange for detailed public disclosure of an invention.
- An invention is a novel solution to a specific technological problem and is a process or a product.
- Patents are a form of **Intellectual Property (IP)**.



The Patent details are:

- Patent application
- Patent specification
- Patent Prosecution
- Patent litigation
- Claims
- Patent pending
- Search
- Examination



Patent application:

- For the grant of a patent, a patent application is filed.
- A patent application is submitted to the patent office.
- It is actually a request pending in the patent office for granting the patent for the invention described and claimed by that application.
- It is also used to refer to the process of applying for a patent, or to the patent specification itself.
- There are three types of patent applications.
 - National
 - Regional
 - International



Patent specification:

- The description of an invention is called as a Patent specification.
- It is submitted together with the official forms and correspondence relating to the application.
- It is a document describing the invention for which the patent is filed and setting out the scope of the protection of the patent.
- A specification generally contains:
 - A title for the application
 - Detailing of the background and overview of the invention
 - A description of the invention
 - Embodiments of the inventions and claims
 - It may include figures to aid the description of the invention



Patent Prosecution:

- It is a process of “negotiating” or “arguing” with a patent office for the grant of a patent.
- It is also the interaction with a patent office with regard to a patent after its grant.

Patent Litigation:

- Patent litigation is related to the legal proceedings for infringement of a patent after it is granted.



Claims :

- Claims define a specific property right.
- The claims must meet relevant requirements for claiming the patent such as novelty, usefulness, and non-obviousness.
- To prevent the misuse of the invention an exclusive right is granted to a patentee in most countries.
- It is the right to prevent others, or at least to try to prevent others, from commercially making, using, selling, importing, or distributing a patented invention without the legal permission.
- The rights of a patent for an invention lies with the first person to file an application for the protection of that invention.



PATENT

Patent Pending :

- It is a warning that an alleged invention is the subject of a patent application.
- This term is used to mark products containing the invention to alert a third party to the fact that the third party may be infringing a patent if the product is copied after the patent is granted.

Search :

- After filing the patent, a search is carried out, either systematically or in some jurisdictions.
- The main purpose of this search is to reveal the prior art which may be relevant to the patentability of the alleged invention.
- The patent is published 18 months after the priority date of the application, and as such is a public document.



PATENT

- The search report is important to check, if there is a prior art which prevents the grant of the useful patent.
- In this case the application may be abandoned before the applicant incurs further expenses.
- Competitors can also use this search report, to get the idea for the scope of protection.



Examination :

- This is a process to ensure that an application complies with the requirements of the relevant patent laws.
- Examination is an iterative process.
- The patent office notifies the applicant about its objection.
- In such a case, the applicant may respond with an amendment or an argument to overcome the objection.
- The amendment or argument may be accepted or rejected.
- After that again the applicant waits for the further response and so forth until a patent is issued or the application is abandoned.



General Patent :

- A general patent format is having following points.
- Similar to a journal research paper, a patent can be read by understanding the following parts.
 - Abstract
 - Background
 - Summary of the invention
 - Detailed description of the invention
 - Claims
 - Figures



CASE STUDIES

- Case study is a method of exploring and analyzing the life of a social unit – be that unit a person, a family, institution, culture group, or even an entire community.
- Case + study:
 - Case is a subject of the inquiry or a bounded system
 - Study refers to developing an in-depth understanding

Objectives of case study

- Clinical purpose – dealing with a patient
- Diagnostic purpose – educational situation to provide the remedial instruction to poor students
- Fact-findings – about psychological or educational problems
- Supplementing other information – It may be a follow up work



Three phases of case study

1. Retrospective phase refers to the past records of the case completely which is used in diagnosing the case
2. Prospective phase refers to the present status of the case, which is helpful in understanding the case. The suggestions and remediation can be offered to the case
3. Conspective phase refers to the future development and improvement of the case which is also employed to examine the effects of the remediation given to the case



Steps involved in case study

- Status of the situation or unit of attention
- Collection of data, examination and history
 - Use multiple sources of evidence\
 - Create a case study database
 - Maintain a chain of evidence
- Diagnosis and identification of casual factors
- Adjustment, treatment and therapy
- Follow up programme



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Thank you